

[Home](#) > [Computational Economics](#) > [Article](#)

Realized Volatility Forecasting for Stocks and Futures Indices with Rolling CEEMDAN and Machine Learning Models

| Published: 12 October 2024

| Volume 66, pages 1215–1268, (2025) [Cite this article](#)

 [Save article](#)

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Accept all cookies

Reject optional cookies

Manage preferences

indices as well as the bean and the metal futures indices. Furthermore, the study employs eight ML methods to predict the decomposed sub-sequences. Finally, the prediction outcomes for the sub-sequences are reconstructed. Through 1-step, 5-step, and 22-step predictions, various evaluation criteria, and the Diebold–Mariano test, the study find that the hybrid CEEMDAN–RF model possesses a better RV prediction ability than the non-hybrid models. Taking the Shanghai Stock Exchange as an example, the mean square error (MSE) evaluation criterion of the RF model is reduced by 52.07% after introducing the CEEMDAN decomposition method. Compared with the other hybrid models, the improvement percentages of the MSE for the hybrid CEEMDAN–RF model are 2.56%, 0.72%, 5.03%, 36.89%, 44.14%, 44.15%, and 58.19%, respectively. The empirical findings indicate that the hybrid CEEMDAN method and ML model can provide accurate, robust results for RV prediction of stocks indices and futures indices.

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to [springer.com](#) and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Buy Now

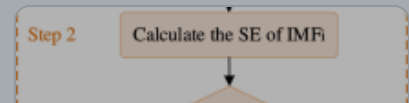
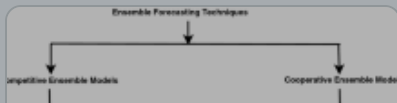
Buy article PDF 39,95 €

Price includes VAT (Sweden)

Instant access to the full article PDF.

[Institutional subscriptions](#) →

Similar content being viewed by others



Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Data Availability

The dataset for the empirical analysis can be derived from WIND, a Bloomberg-like service company in Mainland China providing financial data and information.

References

Andersen, T. G., & Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4), 885–905.
<https://doi.org/10.2307/2527343>

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. [https://doi.org/10.1016/0304-4076\(86\)90063-1](https://doi.org/10.1016/0304-4076(86)90063-1)

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of*

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Fałdziński, M., Fiszeder, P., & Orzeszko, W. (2021). Forecasting volatility of energy commodities: Comparison of garch models with support vector regression. *Energies*, 14(1), 6. <https://doi.org/10.3390/en14010006>

Fang, B., & Ma, S. (2009). Application of BP neural Network in stock market. *Advances in Neural Networks-ISNN*, 2009(5553), 1082–1088. https://doi.org/10.1007/978-3-642-01513-7_119

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Huang, N. E., Shen, Z., Long, S. R., Wu, M. C., Snin, H. H., Zheng, Q., Yen, N. C., Tung, C. C., & Liu, H. H. (1998). The empirical mode decomposition and the Hubert spectrum for nonlinear and non-stationary time series analysis. *Proceedings of the Royal Society a: Mathematical, Physical and Engineering Sciences*, 454(1971), 903–995.

<https://doi.org/10.1098/rspa.1998.0193>

Izzeldin, M., Hassan, M. K., Pappas, V., & Tsionas, M. (2019). Forecasting realised volatility using ARFIMA and HAR models. *Quantitative Finance*, 19(10), 1627–1638.

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Leo Breiman, B., & Eiman, L. B. (1994). Bagging predictors. *Machine Learning*, 24(2), 123–140. <https://doi.org/10.1023/A:1018054314350>

Li, C., Tao, Y., Ao, W., Yang, S., & Bai, Y. (2018). Improving forecasting accuracy of daily enterprise electricity consumption using a random forest based on ensemble empirical mode decomposition. *Energy*, 165, 1220–1227. <https://doi.org/10.1016/j.energy.2018.10.113>

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Liu, C., Hu, Z., Li, Y., & Liu, S. (2017). Forecasting copper prices by decision tree learning. *Resources Policy*, 52, 427–434. <https://doi.org/10.1016/j.resourpol.2017.05.007>

[Article](#) [Google Scholar](#)

Mahkya, D. A., Notodiputro, K. A., & Sartono, B. (2022). Extra trees method for stock price forecasting with rolling origin accuracy evaluation. *Media Statistika*, 15(1), 36–47. <https://doi.org/10.14710/medstat.15.1.36-47>

[Article](#) [Google Scholar](#)

Nelson, D. B. (1991). Conditional heteroscedasticity in asset returns: A new approach. *Econometrica*, 59, 347–370. <https://doi.org/10.2307/2938260>

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Sarigöl, M., & Katipoğlu, O. M. (2023). Estimation of monthly evaporation values using gradient boosting machines and mode decomposition techniques in the southeast anatolia project (GAP) area in Turkey. *Acta Geophysica, Online*. <https://doi.org/10.1007/s11600-023-01067-8>

[Article](#) [Google Scholar](#)

Song, Y., Huang, J., Xu, Y., Ruan, J., & Zhu, M. (2024). Multi-decomposition in deep learning models for futures price prediction. *Expert Systems with Applications*, 246, 123171. <https://doi.org/10.1016/j.eswa.2024.123171>

[Article](#) [Google Scholar](#)

Torres, M. E., Colominas, M. A., Schlotthauer, G., & Flandrin, P. (2011). A complete

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Wu, D., Huang, J. B., & Zhong, M. R. (2020). Prediction of metal futures price volatility and empirical analysis based on symbolic time series of high-frequency. *Transactions of Nonferrous Metals Society of China (English Edition)*, 30(6), 1707–1716.

[https://doi.org/10.1016/S1003-6326\(20\)65332-8](https://doi.org/10.1016/S1003-6326(20)65332-8)

[Article](#) [Google Scholar](#)

Wu, Z., & Huang, N. E. (2008). Ensemble empirical mode decomposition: A noise-assisted data analysis method. *Advances in Adaptive Data Analysis*, 1(1), 1–41.

<https://doi.org/10.1142/S1793536909000047>

[Article](#) [Google Scholar](#)

Yoon, J. (2021). Forecasting of real GDP growth using machine learning models: Gradient

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Funding

Shanghai Planning Project of Philosophy and Social Science, 2023BJB009, Yuping Song.

Author information

Authors and Affiliations

School of Mathematics, Shandong University, Jinan, China

Yuetong Zhang

Zhongtai Securities Institute for Financial Studies, Shandong University, Jinan, China

Ying Peng

School of Finance and Business, Shanghai Normal University, Shanghai, China

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to [springer.com](#) and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Additional information

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Co-first author: Ying Peng.

Appendix

Table [14](#), [15](#), [16](#), [17](#), [18](#), [19](#), [20](#) and [21](#) shows the predictive evaluation results of benchmark model HAR and eight single machine learning models (RF, GB, ET, BAG, BP, KNN, DT, SVR) and eight CEEMDAN method hybrid machine learning models (CEEMDAN-RF, CEEMDAN-

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Table 15 Evaluation results and ranking of realized volatility prediction by different models for futures indexes

Table 16 Performance improvement percentage of added decomposition for stocks indexes

Table 17 Performance improvement percentage of added decomposition for futures indexes

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to [springer.com](#) and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Rights and permissions

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

[Reprints and permissions](#)

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

JEL Classification

[G17](#)

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to [springer.com](#) and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**